

1. Dark image - Grey level Transformation.

Technique: Log Transformation

Justification: A dark image contains most pixel values in low-intensity range, making details difficult to see. Log Transformation expands low-intensity values while compressing high-intensity values.

Formula:

$$S = c \log(1+r)$$

where, r = input pixel intensity

S = output intensity

c = scaling constant.

Working:

→ Increase the brightness of dark pixels.

→ Small intensity values are stretched.

→ Bright regions are compressed highly.

Effect:

→ Hidden details in dark areas become visible.

→ Small intensity values are stretched.

2. Poor Contrast - Histogram

Technique: Histogram Stretching

Justification: When pixel value occupy only a small portion of the intensity range, the image appears dull. Contrast stretching expands these values to use the full intensity range.

Working:

- Finds minimum and maximum intensity values.
- Maps them to the full range (0-255)
- Intermediate values are scaled proportionally.

Effect:

- Improves contrast.
- Makes edges and objects clearer.
- Preserves original brightness relationship.

3. Uniform Intensity Distribution.

Technique - Histogram Equalization

Justification - Histogram equalization redistributes intensity values so that the histogram becomes approximately uniform.

Working process:

1. Calculate the image histogram.
2. Compute the cumulative distribution function (CDF)
3. Map old intensity values to new values using the CDF.
4. Generate enhanced image.

Effects on intensity levels:

- Uses the complete intensity range.
- Increases global contrast.
- Reveals hidden details in both dark and bright regions.

4. Uneven. Brightness Across Regions.

Technique - Adaptive Histogram Equalization.

Justification: Normal histogram equalization enhances the entire image uniformly, but uneven illumination requires local enhancement. CLAHE improve contrast in small regions separately.

Working:

- Divide the image into small blocks.
- Apply histogram equalization to each block.
- Limit excessive contrast.
- Combine neighboring blocks smoothly.

Effect:

- Corrects non-uniform illumination.
- Enhances local details.
- Prevents over-enhancement.

5. Noisy image - Smoothing filter.

Technique - Median filter.

Justification - A median filter is highly effective for removing impulse noise while preserving image edges.

Working mechanism;

- 1) Select a small neighbourhood.
- 2) Arrange the pixel values in ascending order.
- 3) Replace the center pixel with median value.
- 4) Repeat for every pixel

Impact on noise reduction:

- Removes random noise reduction.
- Preserves sharp edges better than averaging filters.
- Produces a smoother image with minimal loss of detail.